Polymer Chemistry

Cite this: *Polym. Chem.*, 2012, **3**, 2006

www.rsc.org/polymers

PAPER

Diffusion ordered spectroscopy (DOSY) as a powerful tool for amphiphilic block copolymer characterization and for critical micelle concentration (CMC) determination

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Received 3rd February 2012, Accepted 29th March 2012
DOI: 10.1039/c2py20054f

Diffusion-ordered spectroscopy (DOSY) NMR was successfully used to characterize amphiphilic block copolymers. A triblock copolymer was prepared by ring-opening polymerisation of a lactide using poly(ethylene glycol) as the initiator. The DOSY NMR experiment is revealed to be a useful analytical method to prove the formation of block copolymers. According to the DOSY map, PLA and PEG blocks exhibited the same diffusion coefficient of 5.623×10^{-10} m^2 s^{-1}, consistent with an efficient polymerisation of the lactide. The determination of the critical micelle concentration (CMC) using DOSY NMR experiments has also been reported. The CMC value correlated with those obtained by fluorimetry and static light scattering. The CMC value was found to be 0.11 g L^{-1}. All the results suggest that DOSY NMR is a valuable analytical tool for the polymer community. The good correlation between the results from DOSY, fluorescence and SLS experiments suggests that the DOSY spectroscopy can accurately measure the CMC of amphiphilic copolymer solutions.

Introduction

In the past decade, amphiphilic block copolymers have attracted considerable interest for their potential applications in medicine, pharmacology, and biotechnology.[1] In particular, polymeric micelles can be used efficiently as drug carriers in drug delivery systems.[2,3] In an aqueous environment, amphiphilic block copolymers are susceptible to form spherical micelles with the core composed of an insoluble hydrophobic block, while the water-soluble hydrophilic block forms the corona or surrounding shell.[4,5] Hydrophobic drugs can be easily encapsulated in the cores of micelles, thus enhancing the drug solubility in aqueous solution.[3]

With the development of living/controlled polymerisation methodologies such as living anionic polymerisation, atom transfer radical polymerisation (ATRP), reversible addition–fragmentation chain transfer polymerisation (RAFT), nitroxide-mediated free radical polymerisation (NMP), ring-opening polymerisation (ROP) or ring-opening metathesis polymerisation (ROMP), amphiphilic block copolymers can be prepared with varying copolymer properties such as chemical composition, molecular weight, and copolymer architecture.[6] Nevertheless, the crude material may contain varying amounts of impurities such as monomers and residual homopolymers as by-products of the synthetic procedure. These heterogeneities negatively affect the final physico-chemical properties of the materials, and purification must be carried out. A number of complementary analytical techniques are available for the characterization of amphiphilic block copolymers. Among them, NMR spectroscopy and liquid chromatography such as size exclusion chromatography (SEC) are probably the most powerful and the most widely used methods due to their versatility. NMR spectroscopy is usually an excellent tool for structural characterization of polymers. Nevertheless, NMR is best suited to the analysis of pure compounds due to the overlapping signals with complex mixtures containing, for example, residual monomers or polymers. On the other hand, SEC is widely used for the determination of the molecular weight and the molecular weight distribution of polymers. Although successful application of SEC with multiple detection has been demonstrated for the block copolymers, limitations have been shown notably in the case of copolymers containing undetectable residual homopolymers.

New characterization methods have been recently used to evaluate the purity of copolymers and the proportion of residual homopolymers.[7] Diffusion ordered NMR spectroscopy (DOSY-NMR) is one of these powerful tools, and has attracted more and more interest in the polymer community in the last few years.[8] DOSY NMR is a two dimensional NMR technique, in which the signal decays exponentially due to the self-diffusion behaviour of individual molecules. This leads to two dimensions: the first

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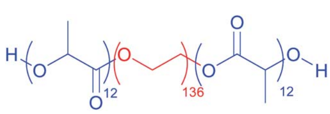
dimension (F_2) accounts for the conventional chemical shift and the second one (F_1) for self-diffusion coefficients (D). The diffusion coefficient is related to the properties of an individual molecule, such as size, mass and charge as well as its surrounding environment, such as solution, temperature and aggregation states. Thus, each component in a mixture can be pseudo-separated, based on its own diffusion coefficient on the diffusion dimension. The advantage of DOSY is that it can be used as a non-invasive method to obtain both physical and chemical information, which constitutes an important alternative to LC-NMR.[9] DOSY was initially invented for the analysis and characterization of mixtures and aggregates.[10] With the development of high field NMR spectrometry, DOSY has also been employed to study the intermolecular interactions in complex systems; a few examples have been described during the last decade.[11–13] For example, DOSY was recently used in polymer chemistry for the structural characterization of daunorubicin-loaded poly(butylcyanoacrylate) nanoparticles.[14] Nevertheless, Chen *et al.* were the first in 1995 to demonstrate that DOSY experiments are useful for polymer chemists to characterize the molecular weight distributions of poly(ethylene oxide) in D_2O.[15] Since then, some applications of DOSY concerning polymer chemistry have been reported in the literature.[16–21] For instance, Jerschow *et al.* used DOSY to study homopolymer mixtures for industrial applications.[16] Recently, block[19,22] and graft[21] copolymers were also characterized by DOSY experiments to elucidate the structure of targeted polymers. Indeed, these results showed that DOSY can be an efficient tool to detect traces of by-products such as impurities, unreacted monomers, residual homopolymers, or degradation products.

This article reports the characterization of amphiphilic block copolymers based on poly(lactic acid) and poly(ethylene glycol) using DOSY NMR spectroscopy. First, the potential of DOSY experiments was confirmed as compared to other techniques for the analysis of block copolymers, especially to determine the formation and the purity of block copolymers. Then, DOSY was used to study the micelle formation and to determine the critical micelle concentration (CMC) of amphiphilic poly(lactic acid)-*block*-poly(ethylene glycol)-*block*-poly(lactic acid) (PLA_{12}–PEG_{136}–PLA_{12}) (Scheme 1). The CMC value determined by DOSY was compared to the CMC value determined by fluorescence and static light scattering (SLS). To the best of our knowledge, no example of CMC determination by DOSY NMR was described in the literature for high molecular weight polymers, although DOSY NMR had been previously used to determine the CMC value of low molecular weight surfactants.[23,24]

Experimental

Materials

All chemicals including deuterated solvents (D_2O and $CDCl_3$) were purchased from Sigma-Aldrich (St-Quentin Fallavier, France). The PLA–PEG triblock copolymer (PLA_{12}–PEG_{136}–PLA_{12}) was prepared by ring-opening polymerisation as previously described.[25]



Scheme 1 Chemical structure of the amphiphilic poly(lactic acid)-*block*-poly(ethylene glycol)-*block*-poly(lactic acid).

NMR study

All ^{1}H NMR and DOSY NMR measurements were performed at 300 K on a Bruker Avance AQS600 NMR spectrometer operating at 600 MHz and equipped with a Bruker multinuclear z-gradient inverse probe head capable of producing gradients in the z direction with strength 55 G cm^{-1}. The DOSY spectra were acquired with the ledbpgp2s pulse program from Bruker topspin software. All spectra were recorded with 32 K time domain data points in the t_2 dimension and 32 t_1 increments. The gradient strength was logarithmically incremented in 32 steps from 2% up to 95% of the maximum gradient strength. All measurements were performed with a compromise diffusion delay Δ of 200 ms in order to keep the relaxation contribution to the signal attenuation constant for all samples. The gradient pulse length δ was 5 ms in order to ensure full signal attenuation. The diffusion dimension of the 2D DOSY spectra was processed by means of the Bruker topspin software (version 2.1). The DOSY maps were obtained with the Bruker topspin software (version 2.1).

The critical micelle concentration (CMC) of the copolymer nanoparticles was determined by 2D DOSY. The micelle copolymer solutions were prepared by direct dissolution of copolymer in deuterated water at different concentrations. The values for the diffusion coefficients D were collected at these different concentrations. The CMC was determined by plotting the diffusion coefficient against the inverse of the polymer concentration. Two slopes are obtained and the CMC was taken as the intersection of these two lines.

Fluorescence measurement

The CMC of the nanoparticles was determined by fluorescence spectroscopy using pyrene as a hydrophobic fluorescent probe. Briefly, an aliquot of pyrene solution (6×10^{-6} M in acetone, 1 mL) was added to different vials, and the solvent was evaporated. Then, 10 mL of aqueous solutions containing different concentrations of copolymer were added to the vials. The final concentration of pyrene in each vial was 6×10^{-7} M. Fluorescence measurements were performed on a RF 5302 Shimadzu spectrofluorometer (Japan) equipped with a xenon light source (UXL-150S, Ushio, Japan). The emission and excitation slit widths were 3 nm and 5 nm, respectively. The samples were excited at 340 nm and emission spectra were recorded from 330 to 440 nm. The emission fluorescence values I_{372} and I_{393}, at 372 and 393 nm, respectively, were used for subsequent calculations. The CMC was determined by plotting the I_{372}/I_{393} ratio against the polymer concentration. The CMC was taken as the intersection of regression lines calculated from the linear portions of the plot.

Static light scattering

Static light scattering (SLS) was carried out at 25 °C with a Malvern Instrument Nano-ZS equipped with a He–Ne laser ($\lambda = 632.8$ nm). The micelle copolymer solutions were prepared by direct dissolution of copolymer in ultrapure water, and were filtered through 0.45 μm PTFE microfilters 24 hours before measurements. The CMC was determined using an autocorrelation function. The correlation function was analyzed *via* the general purpose method (NNLS) to obtain the distribution of diffusion coefficients (D) of the solutes. The scattering intensity proportional to the number and the size of objects present in the solution was recorded. The CMC was determined by plotting the scattering intensity against the polymer concentration. The CMC was taken as the intersection of regression lines calculated from the linear portions of the graph.

Results and discussion

In this study, a triblock copolymer based on poly(lactic acid) (PLA) and poly(ethylene glycol) (PEG) was selected as an example of amphiphilic copolymers able to form micelles in an aqueous environment. The copolymer was prepared by ring-opening polymerisation (ROP) of a lactide using a difunctional poly(ethylene glycol) (M_n = 6000 g mol^{-1}) as the macroinitiator. The molecular weight of the resulting copolymer was determined by ^{1}H NMR as previously reported in the literature,[26,27] and was equal to 7700 g mol^{-1}. The degrees of polymerisation of poly(lactic acid) block and poly(ethylene glycol) block were 24 and 136, respectively. The triblock copolymer, namely PLA_{12}–PEG_{136}–PLA_{12}, was then characterized by DOSY NMR in order to control the efficiency of the copolymerisation. In fact, DOSY NMR can provide evidence of block copolymer formation. In the case of a successful block copolymerisation through, for instance, the polymerisation of monomer B using poly(A) as initiator to prepare poly(A)-*block*-poly(B), the correlation spots characterizing the ^{1}H NMR signals due to poly(A) and poly(B) blocks are characterized by the same diffusion coefficient.

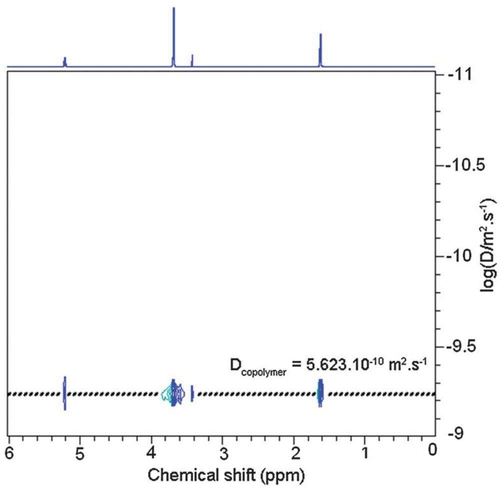


Fig. 1 600 MHz 2D DOSY NMR spectra obtained at 298 K in $CDCl_3$ solution of the copolymer PLA_{12}–PEG_{136}–PLA_{12}.

DOSY NMR of PLA_{12}–PEG_{136}–PLA_{12} was achieved in dilute $CDCl_3$ (Fig. 1). The DOSY map shows an expanded view of the DOSY experiment with the related ^{1}H spectrum projected on top. The ^{1}H NMR spectrum exhibits signals corresponding to the PLA block (δ = 5.15 and 1.56 ppm) and the PEG block (δ = 3.62 ppm). As shown on the DOSY map, ^{1}H NMR signals of PLA and PEG present the same diffusion coefficient 5.623 × 10^{-10} m^2 s^{-1}, consistent with an efficient polymerisation of the lactide on the PEG macroinitiator. To be sure that no residual PEG remains or no PLA homopolymer was formed during the ROP process, dilute $CDCl_3$ solutions of PEG (M_n = 6000 g mol^{-1}) and PLA (M_n = 2100 g mol^{-1}) were analysed by DOSY NMR and were compared to the PLA_{12}–PEG_{136}–PLA_{12} copolymer (Fig. 2A and B). The diffusion coefficients of PEG and PLA were found to be $D = 6.31 \times 10^{-11}$ m^2 s^{-1} and $D = 1.259 \times 10^{-10}$ m^2 s^{-1}, respectively.

Therefore, one can conclude that no residual PEG or PLA homopolymers were observed on the DOSY map of the copolymer (Fig. 1), meaning that all the PEG macroinitiator was consumed during the initiation step of the ROP and no PLA homopolymer was formed during the ROP. It is noteworthy that

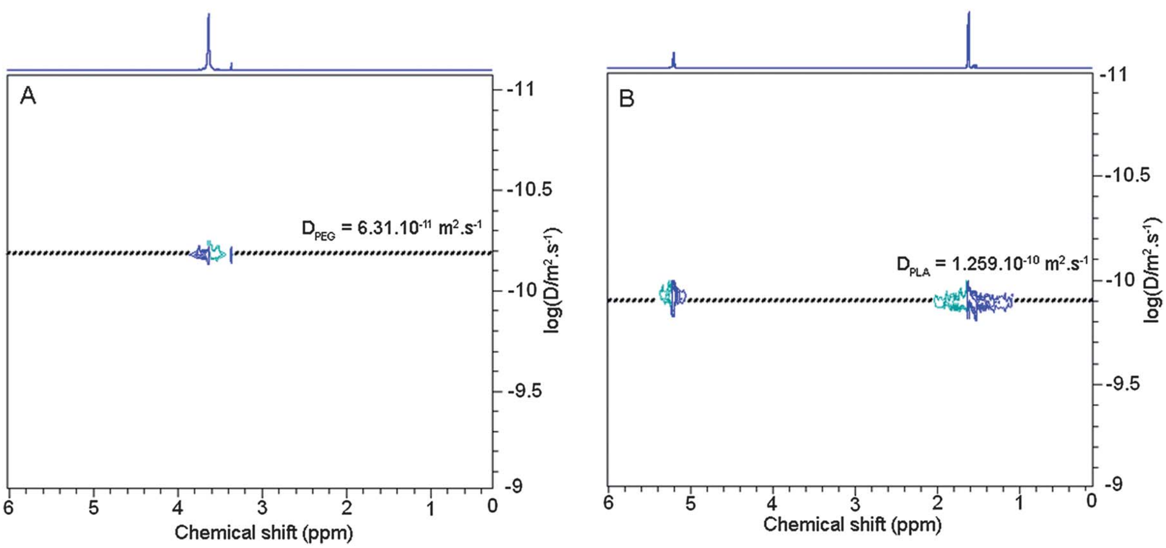


Fig. 2 600 MHz 2D DOSY NMR spectra obtained at 298 K in $CDCl_3$ solution of (A) poly(ethylene glycol), M_n = 6000 g mol^{-1} and (B) poly(lactic acid), M_n = 2100 g mol^{-1}.

DOSY NMR is a very sensitive technique capable of detecting traces in complex mixtures. This method, sometimes called "NMR chromatography", appears complementary to SEC chromatography.

Investigations concerning the formation of micelles in water were then carried out using the DOSY NMR experiment. Fig. 3 shows the DOSY NMR of the PLA_{12}–PEG_{136}–PLA_{12} copolymer in D_2O at a concentration of 5 g L^{-1}. D_2O was expected to preferentially solubilize the hydrophilic PEG part of the copolymer. Indeed, as shown on the DOSY map, only the spot corresponding to PEG block was visible, whereas no spot corresponding to the hydrophobic PLA block was observed, due to the limited molecular motion of the moiety in aqueous solvent. This phenomenon can be interpreted by the formation of micelles or aggregates where the hydrophobic PLA part of the copolymer forms the core and the hydrophilic PEG part forms the shell.

The critical micelle concentration (CMC) of the amphiphilic PLA_{12}–PEG_{136}–PLA_{12} copolymer was determined using DOSY NMR experiments. For this purpose, a series of copolymer solutions in D_2O were prepared with different concentrations, ranging from 1×10^{-2} to 0.6 mg mL^{-1}. DOSY NMR experiments were carried out for each concentration. The CMC was determined by plotting the diffusion coefficient D against the inverse concentration of the copolymer. Two slopes corresponding to the monomeric and micellar aggregation states of the copolymer were obtained and the CMC value was taken as the intersection of these two lines (Fig. 4A). Fig. 4A shows that the diffusion coefficient decreases with increasing copolymer concentration in good agreement with organization of the micelles. The CMC value was found to be 0.11 g L^{-1}. Comparison was then made with CMC values determined by using two other methods: fluorimetry and static light scattering (SLS). The polymer concentration varied from 1×10^{-2} to 0.6 mg mL^{-1} in both experiments. Fluorimetry was achieved in emission with an excitation wavelength at 340 nm. Pyrene was used as a fluorescent probe with a concentration equal to 6×10^{-7} mol L^{-1}. The intensity ratio (I_{372}/I_{393}) of pyrene emission spectra was plotted against the copolymer concentration (Fig. 4B). The CMC was determined as the intersection of regression lines calculated from the two linear portions of the plot. In the case of SLS experiments, the evolution of the scattering intensity as a function of concentration allowed us to determine the CMC as the intensity dramatically increased when micelles were formed in the solution. The CMC values of the block copolymer were found to be 0.09 g L^{-1} and 0.12 g L^{-1} from fluorescence and SLS measurements, respectively, which were very close to that determined by DOSY NMR. These results prove that DOSY NMR is an efficient tool to determine the CMC of amphiphilic copolymers as it is fast, cheap and requires only small amounts of

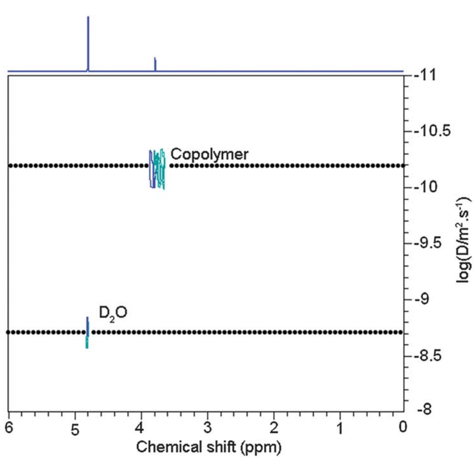


Fig. 3 600 MHz 2D DOSY NMR spectrum recorded at 298 K in D_2O of the copolymer PLA_{12}–PEG_{136}–PLA_{12}.

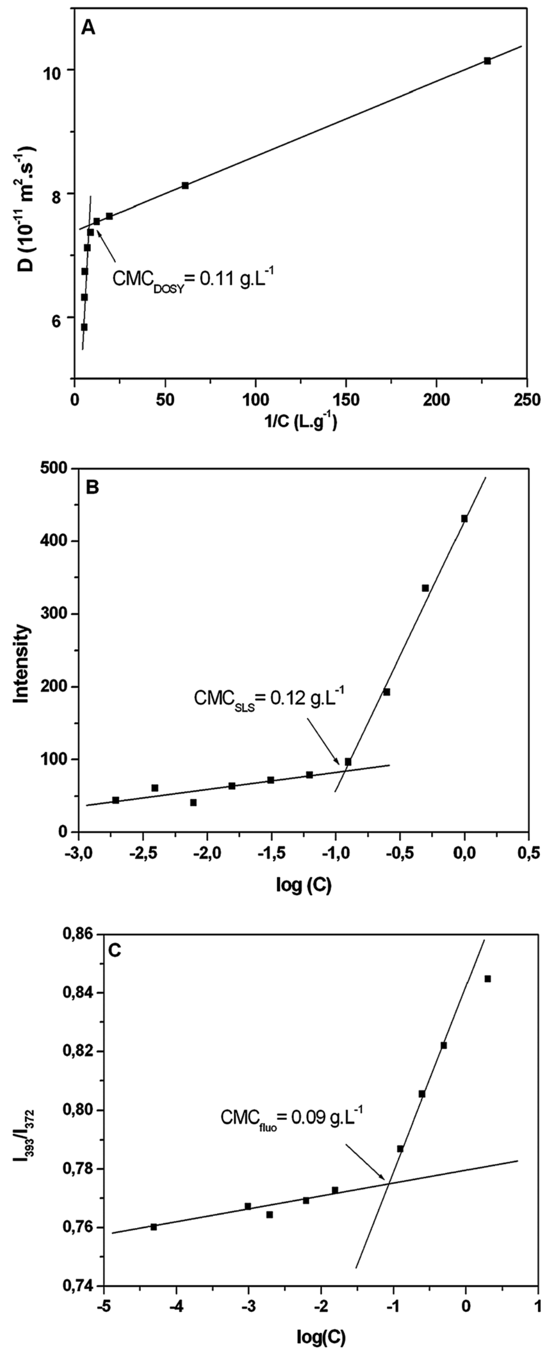


Fig. 4 Determination of the CMC by (A) DOSY NMR, (B) static light scattering, and (C) fluorimetry measurements.

sample. Indeed, to our knowledge, this is the first report on the CMC determination of synthetic polymers by DOSY NMR.

Conclusions

DOSY NMR has been successfully applied to study the structure and the composition of an amphiphilic triblock copolymer based on poly(lactic acid) and poly(ethylene glycol). The formation of a triblock structure was proven by DOSY NMR experiments. Then, the critical micelle concentration (CMC) of this copolymer was determined by different techniques including DOSY NMR, fluorimetry, and SLS. Similar CMC values were obtained in all cases. Therefore, one can conclude that DOSY NMR is an efficient and accurate method to determine the CMC of amphiphilic block copolymers, although this technique is obviously complementary and is not an alternative to other techniques of characterization such as ^{1}H or ^{13}C NMR spectroscopy, SEC chromatography, or MALDI-TOF spectrometry.

Acknowledgements

The authors wish to thank the Erasmus Mundus Action 2 programme (JOSYLEEN) for financial support, and Dr Sophie Monge of the Institut Charles Gerhardt (UMR 5253) for DLS measurements.

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